Practice 1

Create a GitHub Repository and Connect It to Visual Studio 2022

Objective

In this practice, you will:

- create an empty repository on GitHub;
- create a C# project in Visual Studio 2022;
- initialize Git locally;
- connect the local repository to GitHub;
- make an initial commit;
- push the project to GitHub;
- troubleshoot a common authentication problem with private repositories.

Part 1 — Create an Empty Repository on GitHub

1. Go to:

```
https://github.com/new
```

2. Enter a repository name, for example:

```
FoodBankInventory
```

3. Optionally enter a description.

4. Choose the repository visibility:

☑ `Private`

5. For this practice, leave these options unchecked:

☐ `Add a README file`

☐ `Add .gitignore`

☐ `Choose a license`

6. Click:

```
Create repository
```

You should now see an empty GitHub repository.

Part 2 — Create a New C# Project in Visual Studio 2022

1. Open **Visual Studio 2022**.
2. Click:

```
Create a new project
```

3. Select:

```
Console App
```

Use the modern **.NET** template, not **Console App (.NET Framework)**.

4. Click **Next**.
5. Enter the project name:

```
FoodBankInventory
```

6. Choose a local folder.
7. Click **Next**.
8. Select a modern .NET version, for example:

```
.NET 8.0
```

9. Click **Create**.
10. Run the project once to make sure it works.

Part 3 — Initialize Git Locally

In Visual Studio:

1. Open:

```
Git → Create Git Repository
```

2. Create a local Git repository for the current project.
3. Open:

```
View → Git Changes
```

You should see the project files listed as new or modified.

Part 4 — Create the First Commit

1. In the **Git Changes** window, enter a meaningful commit message:

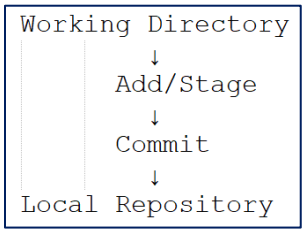
```
Initial commit
```

2. Click:

```
Commit All
```

At this point, the project is recorded in your **local Git repository**.

Conceptually:



The project is **not yet on GitHub**.

Part 5 — Copy the GitHub Repository URL

Go back to the repository page on GitHub.

1. Click:

```
Code
```

2. Select:

```
HTTPS
```

3. Copy the repository URL.

For example:

```
https://github.com/your-username/FoodBankInventory.git
```

Part 6 — Add GitHub as the Remote Repository

Back in Visual Studio:

1. Go to:

```
Tools → Options
```

2. Open:

```
Source Control
  → Git Repository Settings
        → Remotes
```

3. Click:

```
Add
```

4. Enter:

```
Name: origin
```

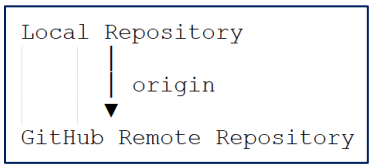
5. Paste the GitHub repository URL into the Fetch and Push URL fields.

For example:

```
https://github.com/your-username/FoodBankInventory.git
```

6. Save the remote settings.

Your setup should now be:



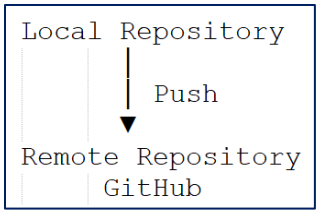
Part 7 — Push the Project to GitHub

1. Open:

```
View → Git Changes
```

2. Click the **Push** icon.

Conceptually:



3. Return to GitHub and refresh the repository page.

You should now see files such as:

`Program.cs`

`FoodBankInventory.csproj`

`FoodBankInventory.sln`

You should also see the commit message:

`Initial commit`

Troubleshooting — "Repository Not Found"

If Visual Studio shows:

`Repository not found`

first verify the remote URL.

Go to:

`Tools`

→ `Options`

→ `Source Control`

→ `Git Repository Settings`

→ `Remotes`

Confirm that `origin` exactly matches the URL shown on GitHub.

For example:

`https://github.com/your-username/FoodBankInventory.git`

If the URL is correct and the repository is **Private**, the problem is often authentication.

Fix Authentication for a Private GitHub Repository

Step 1 — Check the Visual Studio Account

In Visual Studio, click your account/profile icon.

Make sure you are signed in with the GitHub account that owns the repository.

Then try **Push** again.

Step 2 — Clear Old GitHub Credentials

If push still fails:

1. Close Visual Studio.
2. Open Windows:

```
Credential Manager
```

3. Select:

```
Windows Credentials
```

4. Look for entries related to:

```
github.com
```

or:

```
git:https://github.com
```

5. Remove the old GitHub credentials.
6. Reopen Visual Studio.
7. Try **Push** again.
8. Git Credential Manager should ask you to sign in.
9. Sign in using the GitHub account that owns the private repository.
10. Authorize access if prompted.

Then push again.

Part 8 — Verify the Final Result

On GitHub, confirm that:

- your project files are visible;
- the repository contains your initial commit;
- the repository is still private if required.

The complete workflow is:

```
1. Create Repository on GitHub
        ↓
2. Create Project in Visual Studio
        ↓
3. Initialize Git
        ↓
4. Commit Locally
        ↓
5. Add Remote "origin"
        ↓
6. Authenticate with GitHub
        ↓
7. Push
        ↓
8. Verify on GitHub
```

Key Concept

Remember:

Commit = *Save changes in the local repository*

Push = *Send committed changes to GitHub*

They are two different operations.